

Introduction to Economics

Addis Ababa University

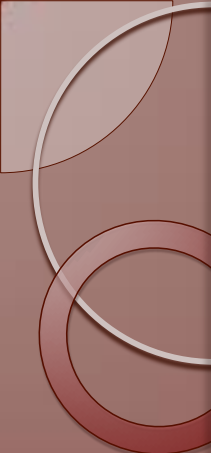
College of Business and Economics

Department of Economics



Chapter Three

Theory of Consumer Behavior



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- Consumer behavior can be best understood in three steps.
 - **Examining consumer's preference**, to describe how people prefer one good to another
 - **Looking at consumers budget constraints** – they have limited incomes that restrict the quantities of goods they can buy.
 - **Putting consumer preference and budget constraint together** to determine consumer choice.

Consumer Preferences and Choices

Axioms of consumer preferences

- **Strict preference:** Given any two consumption bundles (X_1, X_2) and (Y_1, Y_2) , if $(X_1, X_2) > (Y_1, Y_2)$
- **Weak preference:** If the consumer is indifferent between the two commodity bundles or if $(X_1, X_2) \geq (Y_1, Y_2)$, the consumer would be equally satisfied if he consumes (X_1, X_2) or (Y_1, Y_2) .
- **Completeness:** For any two commodity bundles X and Y , a consumer will prefer X to Y , Y to X or will be indifferent between the two.
- **Transitivity and Consistency:**
- **More is better than less:** Consumers always prefer more of any good to less and they are never satisfied or satiated.

UTILITY

- Utility is the level of satisfaction that is obtained by consuming a commodity or undertaking an activity.
- Utility is characterized by the following
 - **‘Utility’ and ‘Usefulness’ are not synonymous.**
 - **Utility is subjective.** The utility of a product will vary from person to person.
 - The utility of a product can be different at different places and time.

Approaches of measuring utility

There are **TWO** Approaches

- I. The cardinal utility theory
- II. The ordinal utility theory

I. The cardinal utility theory

- According to the cardinal utility theory, utility is measurable by arbitrary unit of measurement called **utils** in the form of 1, 2, 3 etc.

I.I. Assumptions of cardinal utility theory

- 1. Rationality of consumers**
- 2. Utility is cardinally measurable:** Utility is measured in objective terms/units called utils.
- 3. Constant marginal utility of money:** A given unit of money deserves the same value at any time or place it is to be spent.
- 4. Diminishing marginal utility (DMU):** This refers to the marginal utility of successive units of consumption
- 5. The total utility** of a basket of goods depends on the quantities of the individual commodities. $TU = f(X_1, X_2, \dots, X_n)$.

1.2. Total and marginal utility

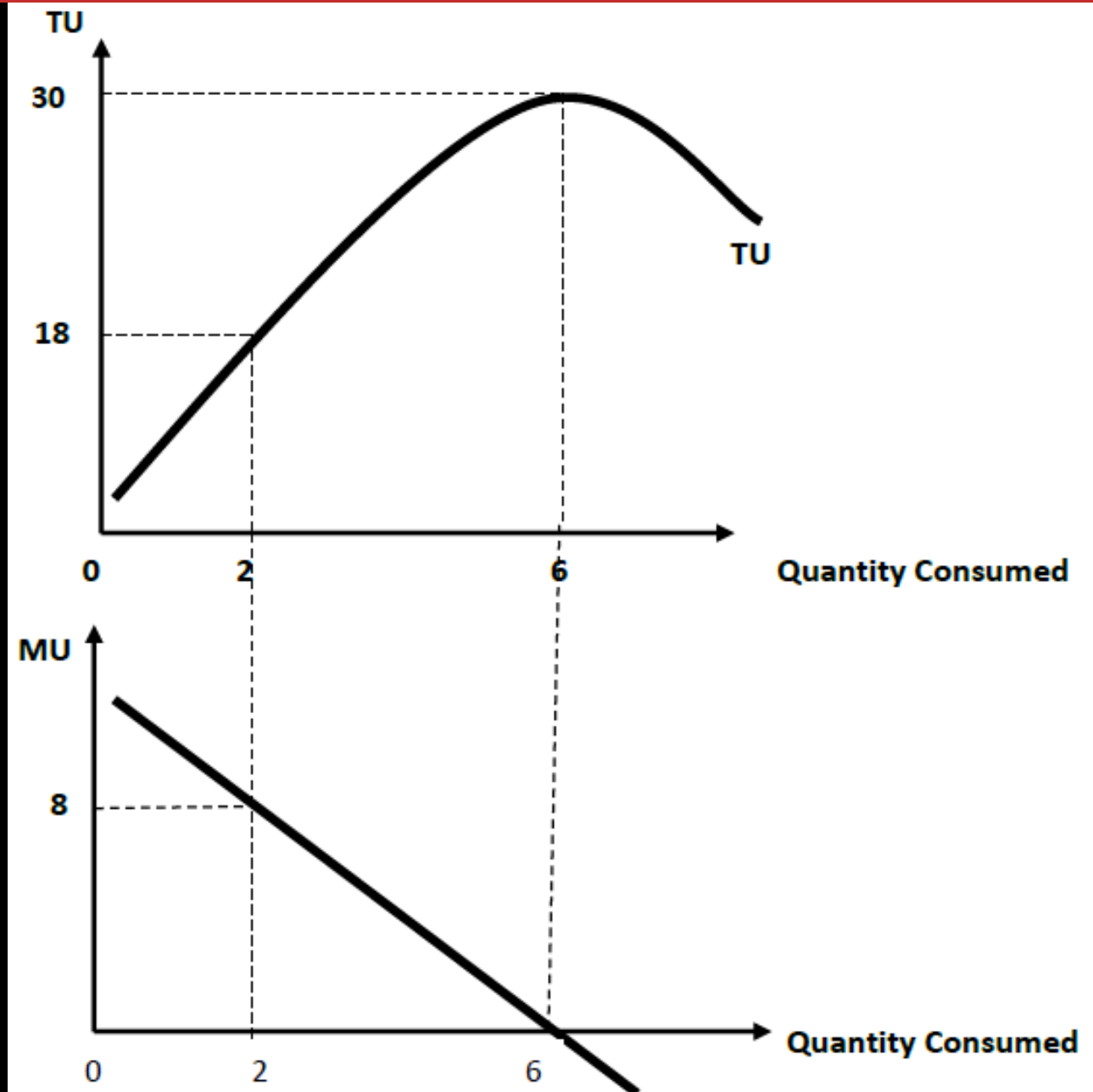
- **Total Utility** (TU) is the total satisfaction a consumer gets from consuming some specific quantities of a commodity at a particular time.
- **Marginal Utility** (MU) is the *extra satisfaction* a consumer realizes from an additional unit of the product.
 - It is the change in total utility that results from the consumption of one more unit of a product.
 - Graphically, MU is the slope of total utility.

- Mathematically, marginal utility is: $MU = \frac{\Delta TU}{\Delta Q}$
where, ΔTU is the change in total utility, and ΔQ is the change in the amount of product consumed.

Table : Total and marginal utility

Quantity	Total utility (TU)	Marginal utility (MU)
0	0	-
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

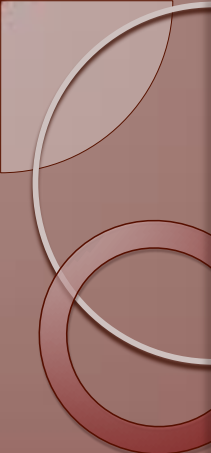
Graphically, the above data can be depicted as follows.



- As it can be seen from the above figure,
 - When TU is increasing, MU is positive.
 - When TU is maximized, MU is zero.
 - When TU is decreasing, MU is negative.

1.3. Law of Diminishing Marginal Utility (LDMU)

- The LDMU states that as the quantity consumed of a commodity increases per unit of time, the utility derived from each successive unit decreases, consumption of all other commodities remaining constant.
- The extra satisfaction that a consumer derives declines as he/she consumes more and more of the product in a given period of time.

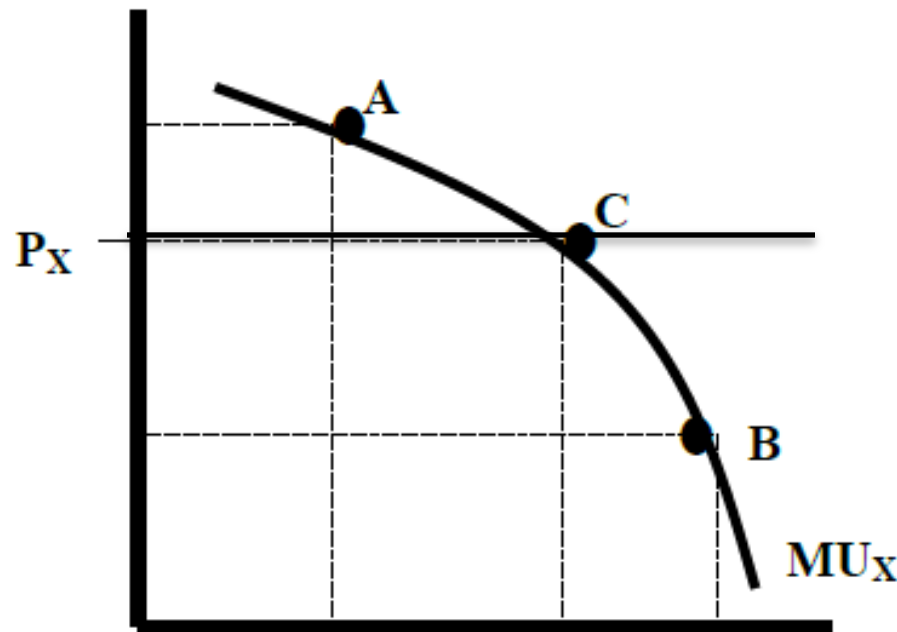
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- The law of diminishing marginal utility is based on the following assumptions.
 - The consumer is rational
 - The consumer consumes identical or homogenous product. The commodity to be consumed should have similar quality, color, design, etc.
 - There is no time gap in consumption of the good
 - The consumer taste/preferences remain unchanged

I.4. Equilibrium of a consumer

- The objective of a rational consumer is to maximize total utility.
- As long as the additional unit consumed brings a positive marginal utility, the consumer wants to consume more of the product because total utility increases.
- Given the limited income and the price level of goods and services, what combination of goods and services should the consumer consume so as to get the maximum utility?
- Two cases of equilibrium
 - I. One commodity case, &**
 - II. Two or more commodity case**

I. The case of one commodity

- The equilibrium condition of a consumer that consumes a single good X occurs when the marginal utility of X is equal to its market price. $MU_X = P_X$



- At any point above point C like point A where **$MU_x > P_x$** , it pays the consumer to consume **more**.
- At any point below point C like point B where **$MU_x < P_x$** , the consumer consumes **less** of X.
- However, at point C where **$MU_x = P_x$** the consumer is at **equilibrium**.

II. Case of two or more commodities

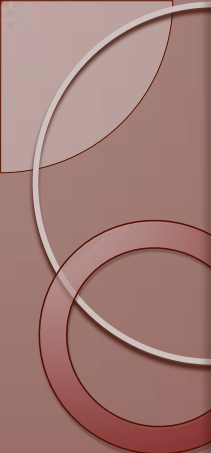
- The consumer's equilibrium is achieved when the marginal utility per money spent is equal for each good purchased and his money income available for the purchase of the goods is exhausted.

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} = \dots = \frac{MU_N}{P_N} \quad \text{and} \quad P_X Q_X + P_Y Q_Y + \dots + P_N Q_N = M$$

where, M is the income of the consumer.

Example: Suppose Saron has 7 Birr to be spent on two goods: banana and bread. The unit price of banana is 1 Birr and the unit price of a loaf of bread is 4 Birr. The total utility she obtains from consumption of each good is given below.

<i>Income = 7 Birr, Price of banana = 1 Birr, Price of bread = 4 Birr</i>							
<i>Banana</i>				<i>Bread</i>			
<i>Quantity</i>	<i>TU</i>	<i>MU</i>	<i>MU/P</i>	<i>Quantity</i>	<i>TU</i>	<i>MU</i>	<i>MU/P</i>
0	0	-	-	0	0	-	-
1	6	6	6	1	12	12	3
2	11	5	5	2	20	8	2
3	14	3	3	3	26	6	1.5
4	16	2	2	4	29	3	0.75
5	16	0	0	5	31	2	0.5
6	14	-2	-2	6	32	1	0.25

- 
- Recall that utility is maximized when the condition of marginal utility of one commodity divided by its market price is equal to the marginal utility of the other commodity divided by its market price.
 - From the table above, there are two different combinations of the two goods where MU of the last birr spent on each commodity is equal.
 - But, only one of the two combinations is consistent with the prices of the goods and her income.
 - Saron will be at equilibrium when she consumes 3 units of banana and 1 loaf of bread.
 - The total utility that Saron derives from this combination 26 ($=14+12$)

2. The ordinal utility theory

- The ordinal utility theory is also known as the *indifference curve approach*.
- It is not possible for consumers to express the utility of various commodities they consume in **absolute terms**, like 1 util, 2 utils, or 3 utils
 - but it is possible to express the utility in **relative terms**.
 - consumers can rank commodities in the order of their preferences as 1st, 2nd, 3rd and so on.
 - Therefore, the consumer need not know in specific units the utility of various commodities to make his choice.

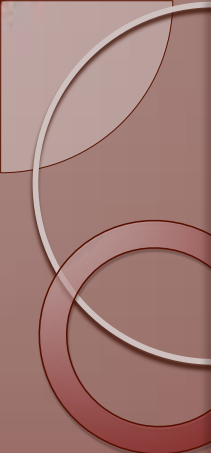
2.1. Assumptions of ordinal utility theory

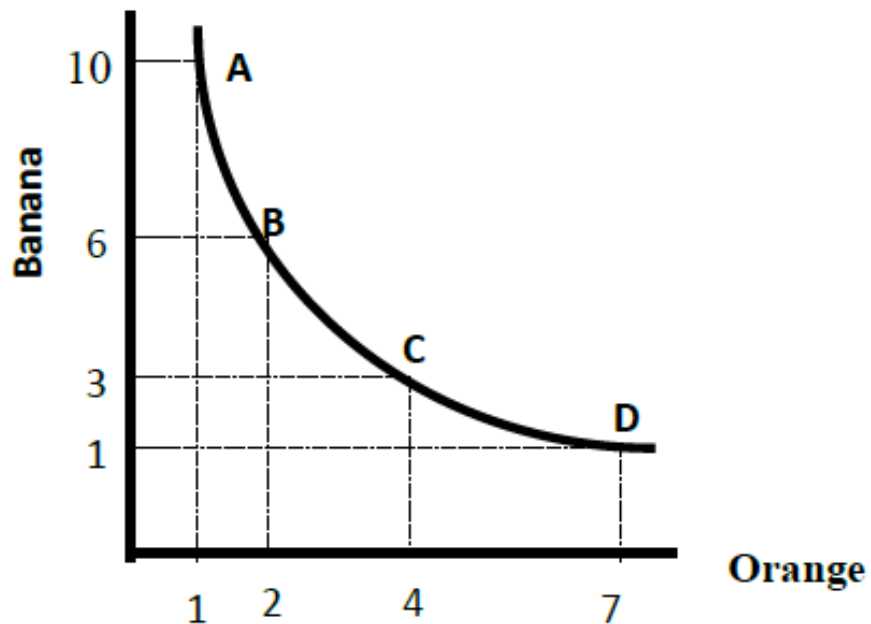
- i. **Consumers are rational**
- ii. **Utility is ordinal** - utility is not absolutely (cardinally) measurable.
- iii. **Diminishing marginal rate of substitution (MRS):** **MRS** is the rate of substitution is the rate at which a consumer is willing to substitute one commodity for another commodity so that his total satisfaction remains the same.
- iv. The **total utility** of a consumer is measured by the quantities of all items he/she consumes from his/her **consumption basket**.
- v. **Consumer's preferences are consistent and transitive.**

2.2. Indifference set, curve and map

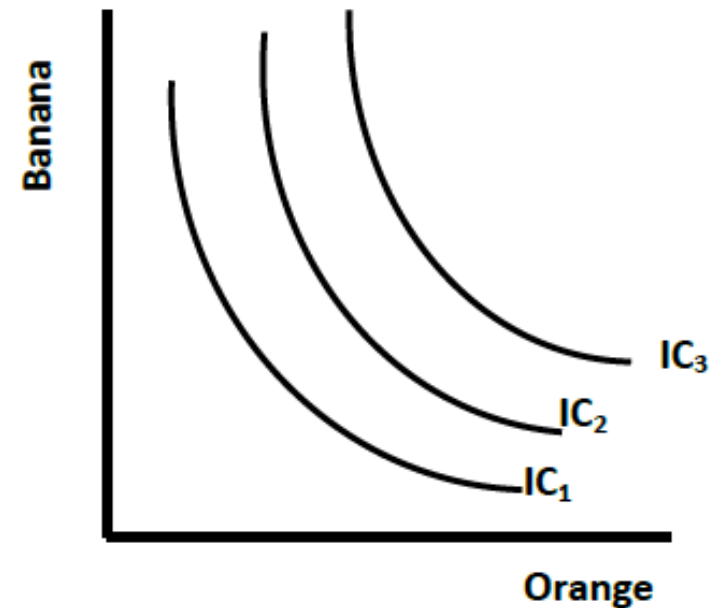
- **Indifference set or schedule** is a combination of goods for which the consumer is indifferent.
 - It shows the various combinations of goods from which the consumer derives the same level of satisfaction.

Bundle (Combination)	A	B	C	D
Orange	1	2	4	7
Banana	10	6	3	1

- 
- When the indifference set/schedule is expressed graphically, it is called an *indifference curve*.
 - **Indifference curve:** An indifference curve shows different combinations of two goods which yield the same utility (level of satisfaction) to the consumer.
 - A group of indifference curves is called **indifference map**.



i) Indifference curve



ii) Indifference map

Bundle (Combination)	A	B	C	D
Orange	1	2	4	7
Banana	10	6	3	1

Properties of indifference curves (ICs)

1. Indifference curves have negative slope

- Implies the consumption one good can be increased only by reducing the consumption of the other.
- To keep the utility constant, as the quantity of one good is increased the other must be decreased.
- Shows trade-off and substitution between goods

2. ICs are convex to the origin.

- implies that slope of an IC decreases as we move along the curve from the left to the right.
- Also, implies that MRS diminishes.
- Commodities can substitute one another at any point on an indifference curve but are **not perfect substitutes**.

3. A higher indifference curve is always preferred to a lower one.

- The further away from the origin an indifferent curve lies, the higher the level of utility it denotes.
- A higher indifference curve contains more of the two commodities than the lower ones.

4. Indifference curves never cross each other (cannot intersect).

- Intersection of ICs rules out the assumptions of consistency and transitivity
- The figure below shows the violations of the assumptions of preferences due to the intersection of indifference curves.

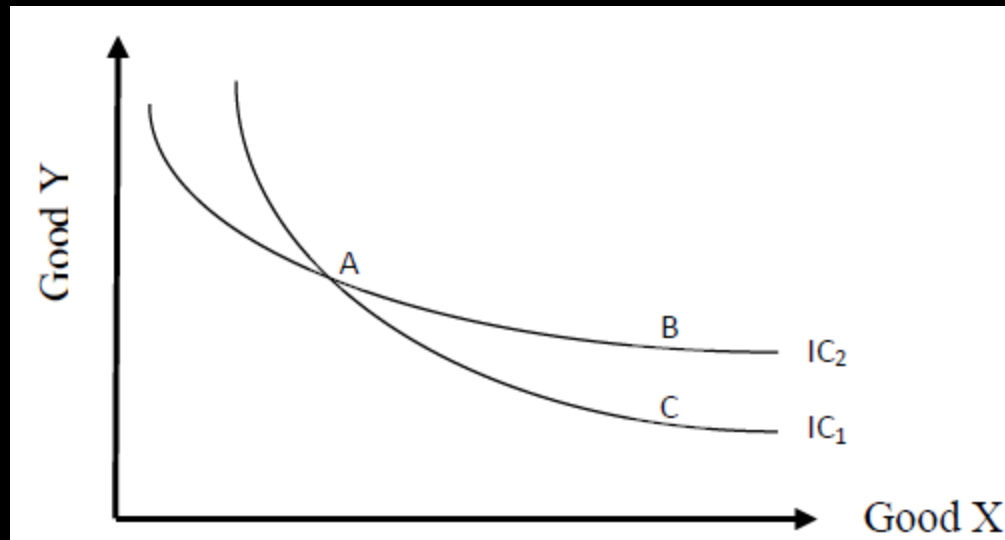


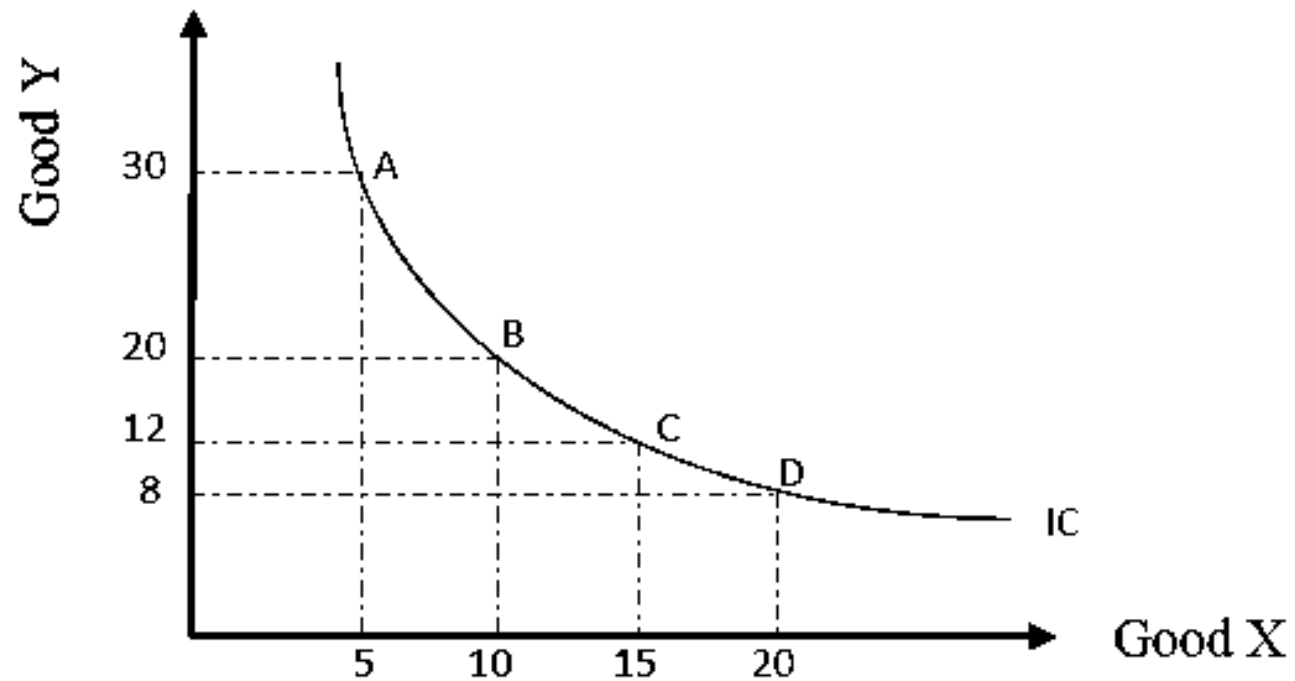
Figure : Intersection of indifference curves

- The figure implies that the consumer prefers bundle B to bundle C.
- On IC_1 , the consumer is indifferent between bundle A and C, and along IC_2 the he is indifferent between bundle A and B.
- By transitivity, this implies that the consumer is indifferent between bundle B and C which is contradictory or inconsistent with the initial statement where the consumer prefers bundle B to C.
- Therefore, indifference curves never cross each other.

Marginal rate of substitution (MRS)

- Marginal rate of substitution is a rate at which consumers are willing to substitute one commodity for another in such a way that the consumer remains on the same indifference curve.
- Marginal rate of substitution of X for Y is defined as the number of units of commodity Y that must be given up in exchange for an extra unit of commodity X so that the consumer maintains the same level of satisfaction.
- Since one of the goods is sacrificed to obtain more of the other good, the MRS is negative.

$$MRS_{X,Y} = \frac{\text{Number of units of } Y \text{ given up}}{\text{Number of units of } X \text{ gained}} = -\frac{\Delta Y}{\Delta X}$$



It is also possible to derive MRS using the concept of marginal utility.

$$MRS_{X,Y} = \frac{MU_X}{MU_Y}$$

- **Proof:** Suppose the utility function for two commodities X and Y is defined as:

$$U = f(X, Y)$$

- Since utility is constant along an indifference curve, the total differential of the utility function will be zero.

$$dU = \frac{\partial U}{\partial X} dX + \frac{\partial U}{\partial Y} dY = 0$$

- From the total differential, we can get

$$MRS_{X,Y} = \frac{MU_X}{MU_Y}$$

- **Example:** Suppose a consumer's utility function is given by $U(X, Y) = X^4 Y^2$

$$MRS_{X,Y} = 2Y/X$$

Quiz (5%)

I. Suppose a particular consumer has 8 birr to be spent on two goods, A and B. The unit price of good A is 2 birr and the unit price of B is 1 birr. The marginal utility (MU) she gets from consumption of the goods is given below.

A) Based on the cardinal analysis, what is the combination of the two goods that gives maximum utility to the consumer?

B) What is the total utility at the utility maximization level?

Quantity	MU_A	MU_B
1	36	30
2	24	22
3	20	16
4	18	12
5	16	10
6	10	4

2.3. The budget line

- It is also called the **price line**
- The *budget line* is a set of the commodity bundles that can be purchased if the entire income is spent.
- It is a graph which shows the various combinations of two goods that a consumer can purchase given his/her limited income and the prices of the two goods.

- Assume that there are two goods (good X and good Y) with their prices Price of good X (P_x) and Price of good Y (P_y) and limited income of the consumer (M).

$$M = P_x X + P_y Y$$

- By rearranging the above equation, we can derive the following general equation of a budget line.

$$Y = \frac{M}{P_y} - \frac{P_x}{P_y} X$$

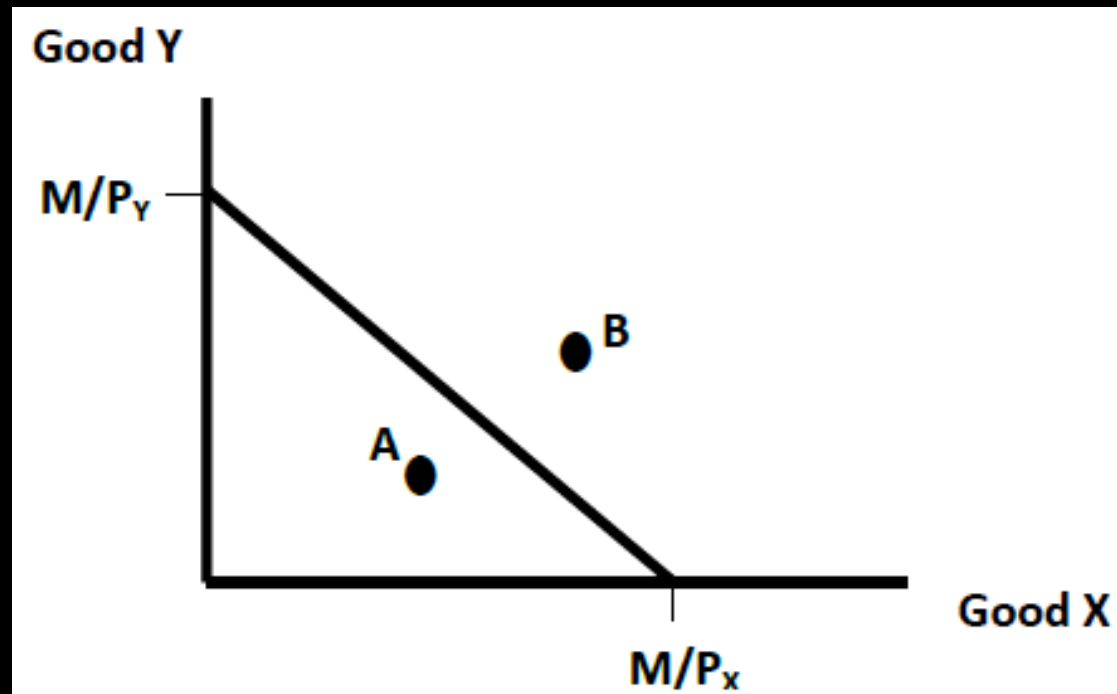


Figure : The budget line

- The slope of the budget line is given by $-P_X/P_Y$
- **Example:** A consumer has \$1000 to spend on two goods X and Y with prices \$10 and \$25 respectively. Derive the equation of the budget line and sketch the graph.

Effects of Change in Income and Prices on the Budget Line

A. Change in income:

- If the income of the consumer changes (keeping the prices of the commodities unchanged), the budget line also shifts (changes).
- Increase in income causes an upward/outward shift in the budget line that allows the consumer to buy more goods and services and
- Decrease in income causes a downward/inward shift in the budget line that leads the consumer to buy less quantity of the two goods.
- It is important to note that the slope of the budget line (the ratio of the two prices) does not change when income rises or falls.

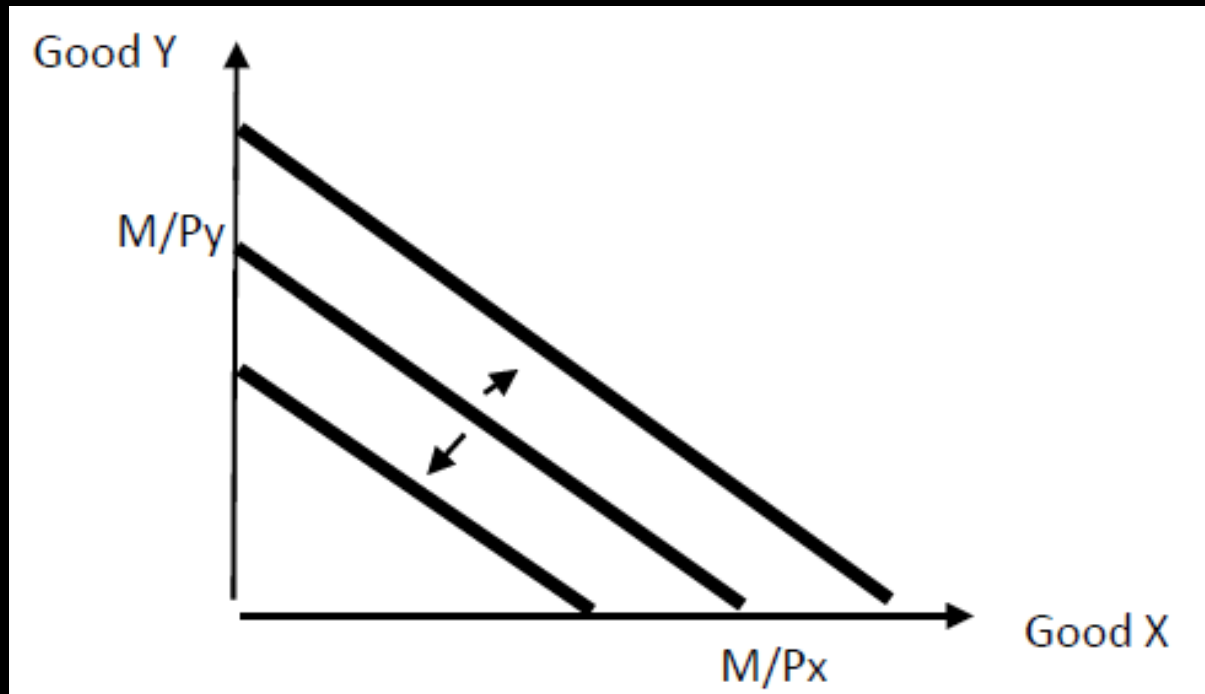


Figure: Effects of increase (right) and decrease (left) in income on the budget line

B. Change in prices:

- An equal increase in the prices of the two goods shifts the budget line inward.
- An equal decrease in the prices of the two goods, on the other hand, shifts the budget line outward.

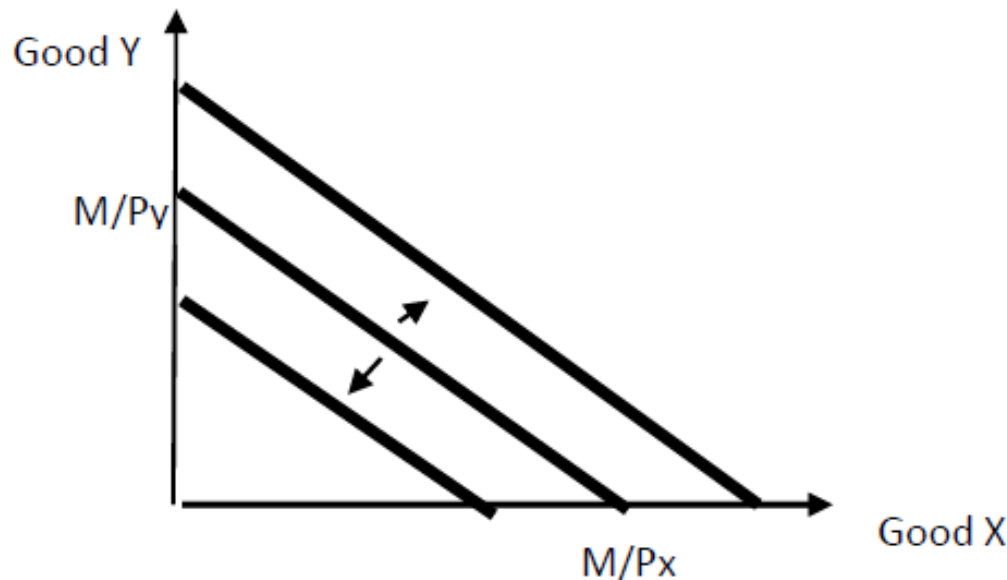


Figure: Effect of proportionate increase (inward) and decrease (outward) in the prices of both goods

- An increase or decrease in the price of one of the two goods, keeping price of the other good & income constant, changes the slope of the budget line by affecting only the intercept of the commodity that records the change in the price.

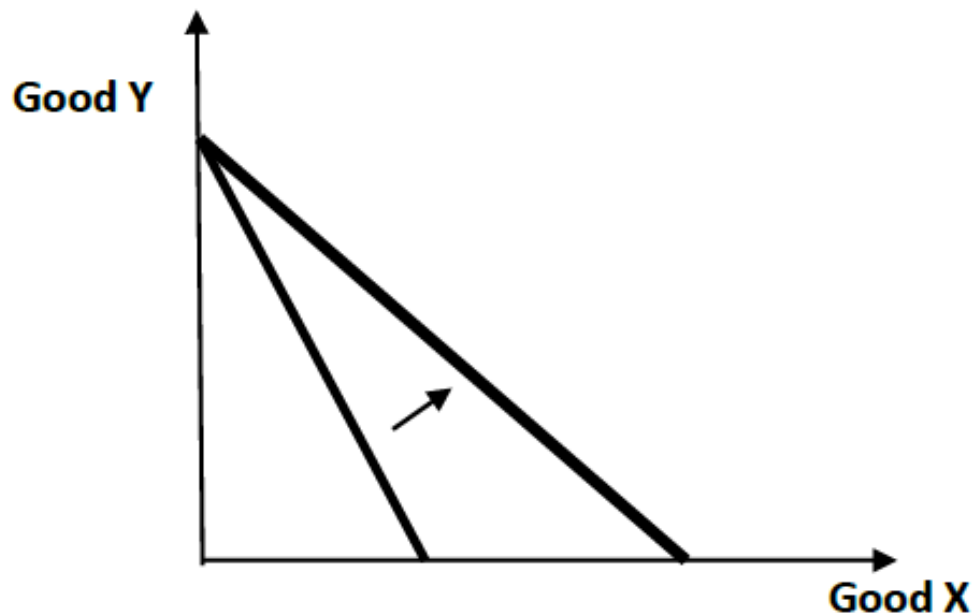


Figure : Effect of decrease in the price of only good X on the budget line

2.4. *Equilibrium of the consumer*

- A rational consumer tries to attain the highest possible indifference curve, given the budget line.
- This occurs at the point where the indifference curve is tangent to the budget line so that the slope of the indifference curve is equal to the slope of the budget line.

- In figure below, the equilibrium of the consumer is at point E where the budget line is tangent to the highest attainable indifference curve (IC_2).

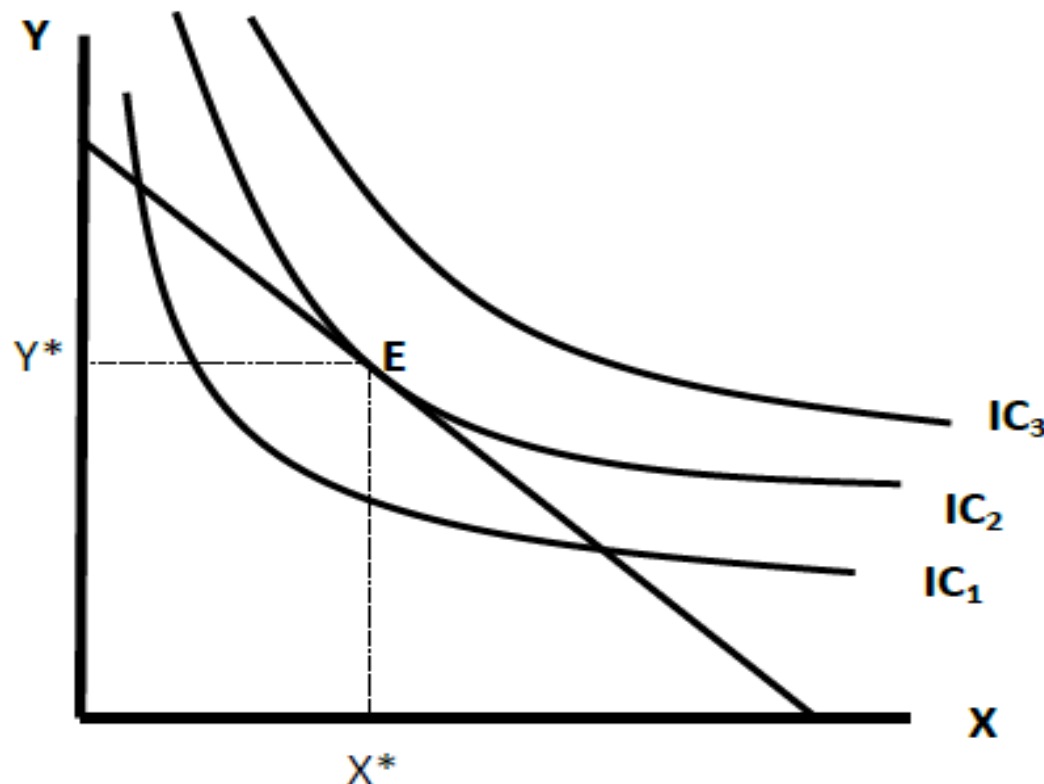


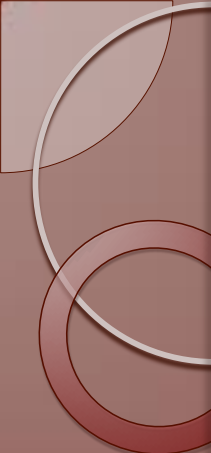
Fig : Consumer equilibrium under indifference curve approach

- Mathematically, consumer optimum (equilibrium) is attained at the point where:

Slope of indifference curve = Slope of the budget line

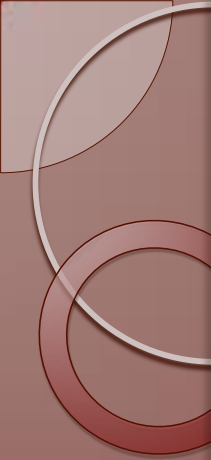
$$MRS_{XY} = \frac{P_X}{P_Y}$$

$$\frac{MU_X}{MU_Y} = \frac{P_X}{P_Y}$$



Example: If she spends all of her income on trouser and shoe during Christmas shopping, Rahel can afford 8 trousers and 8 pairs of shoes. She could also use her entire budget to buy 16 trousers and 4 pairs of shoes. If the price of a trouser is 400 Birr, then

- A) How much is Rahel's income allocated for Christmas shopping?
- B) What is the budget line equation of Rahel?
- C) If Rahel's utility function is $U = T^{0.8}S^{0.2}$, where T is Trousers and S is Pair of Shoes. What is the optimal combination of Trousers and Pair of Shoes for Rahel?



End of Chapter Three